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Gramlich et al.

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(54) **DYNAMIC SLAB GATE VALVES**

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F16K 3/20 (2006.01)
F16K 3/02 (2006.01)
F16K 3/36 (2006.01)

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CPC ***F16K 3/20*** (2013.01); ***F16K 3/0227*** (2013.01); ***F16K 3/36*** (2013.01)

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CPC F16K 3/20; F16K 3/0227; F16K 3/36
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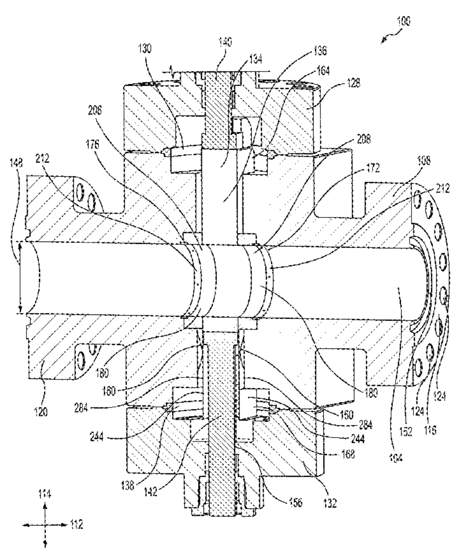
(57) **ABSTRACT**

A dynamic slab gate valve for use in controlling fluid flow through pipelines and other tubulars has a dynamic seat assembly and a dynamic skirt assembly. The dynamic seat assembly is urged toward a gate of the dynamic slab gate valve by an energizer to maintain a seal between a seat and the gate. The dynamic skirt assembly is urged toward the gate of the dynamic slab gate valve by an energizer to maintain a seal with the gate.

26 Claims, 12 Drawing Sheets



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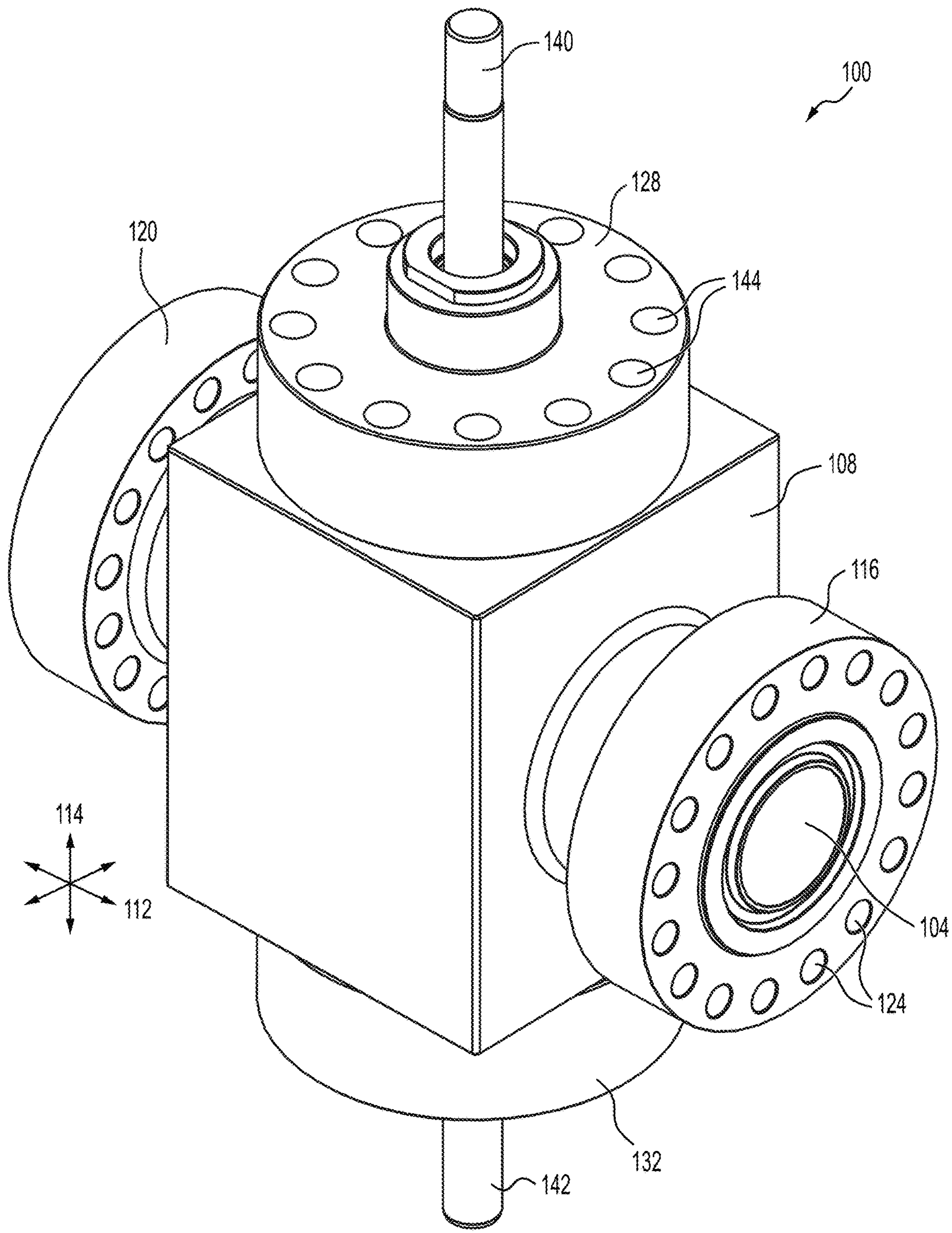

FIG. 1

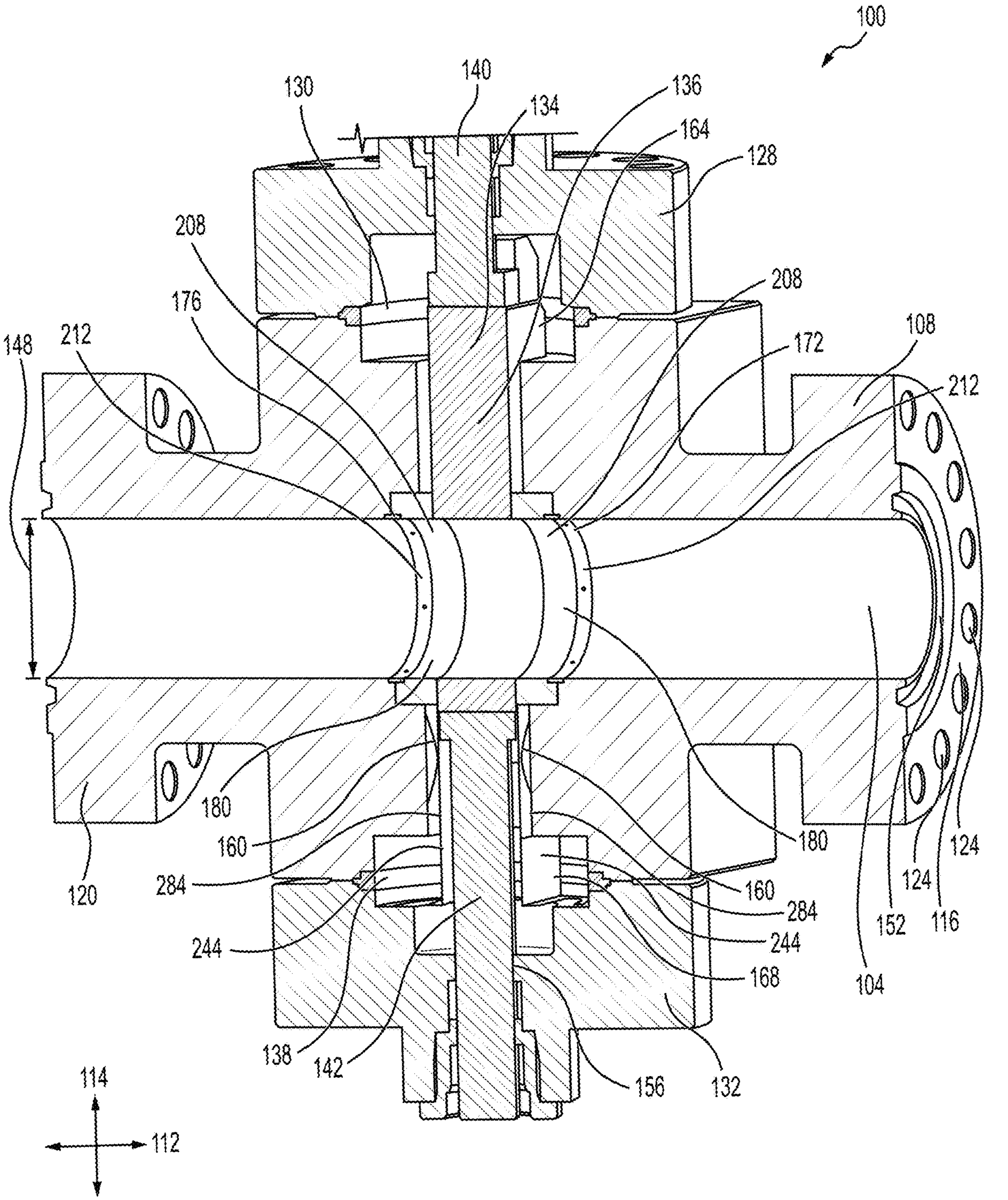
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FIG. 2

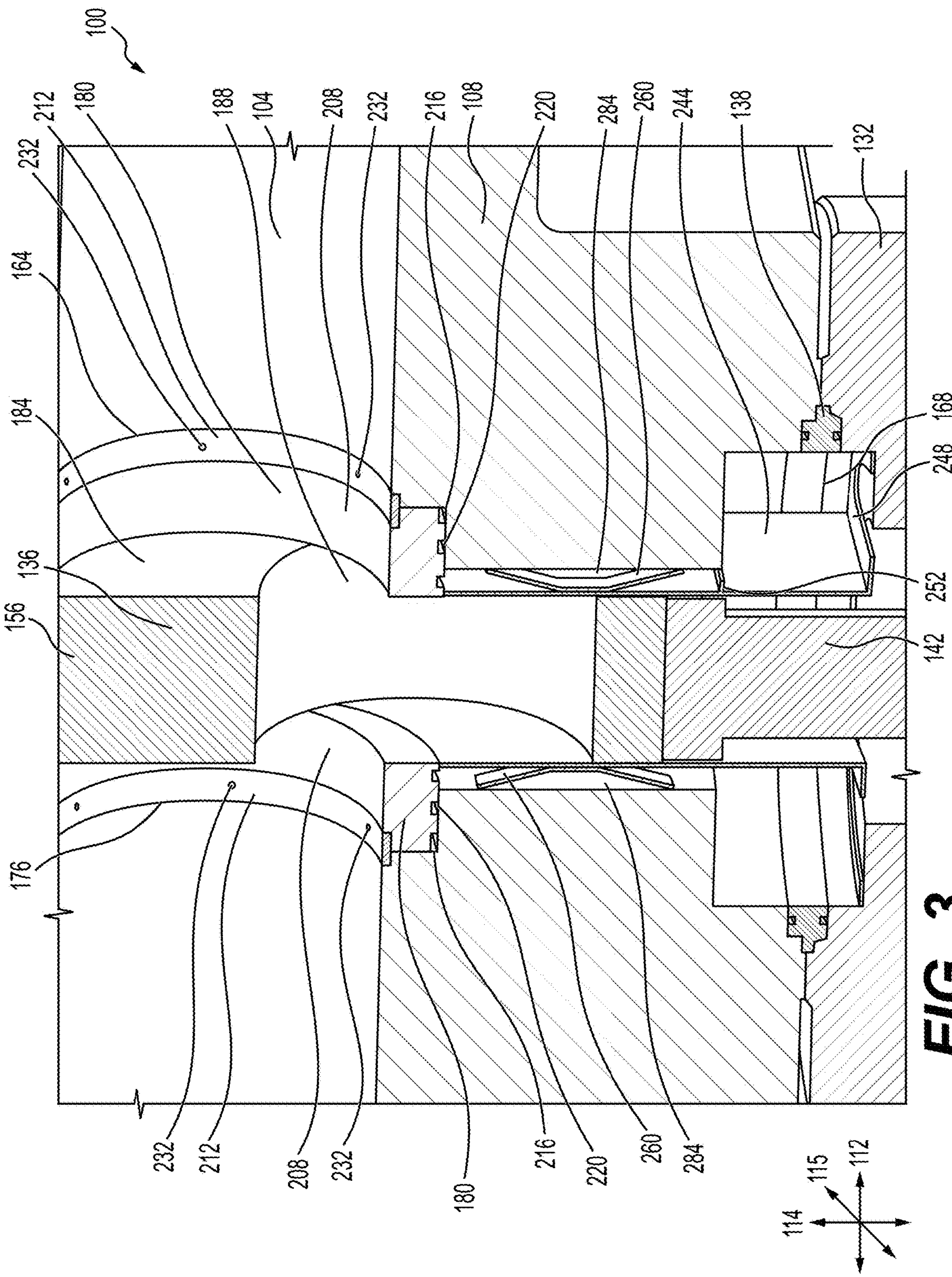
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FIG. 3

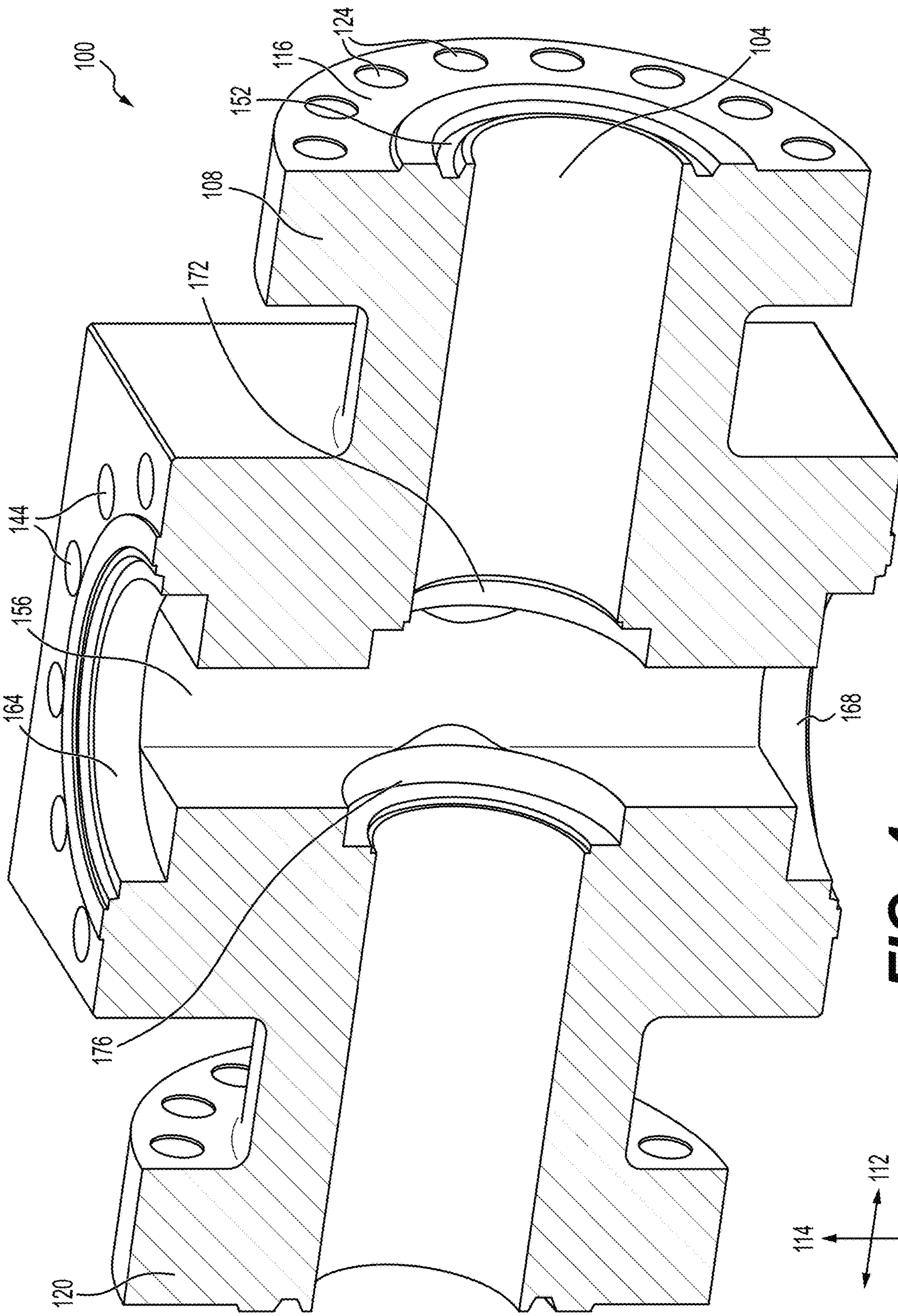


FIG. 4

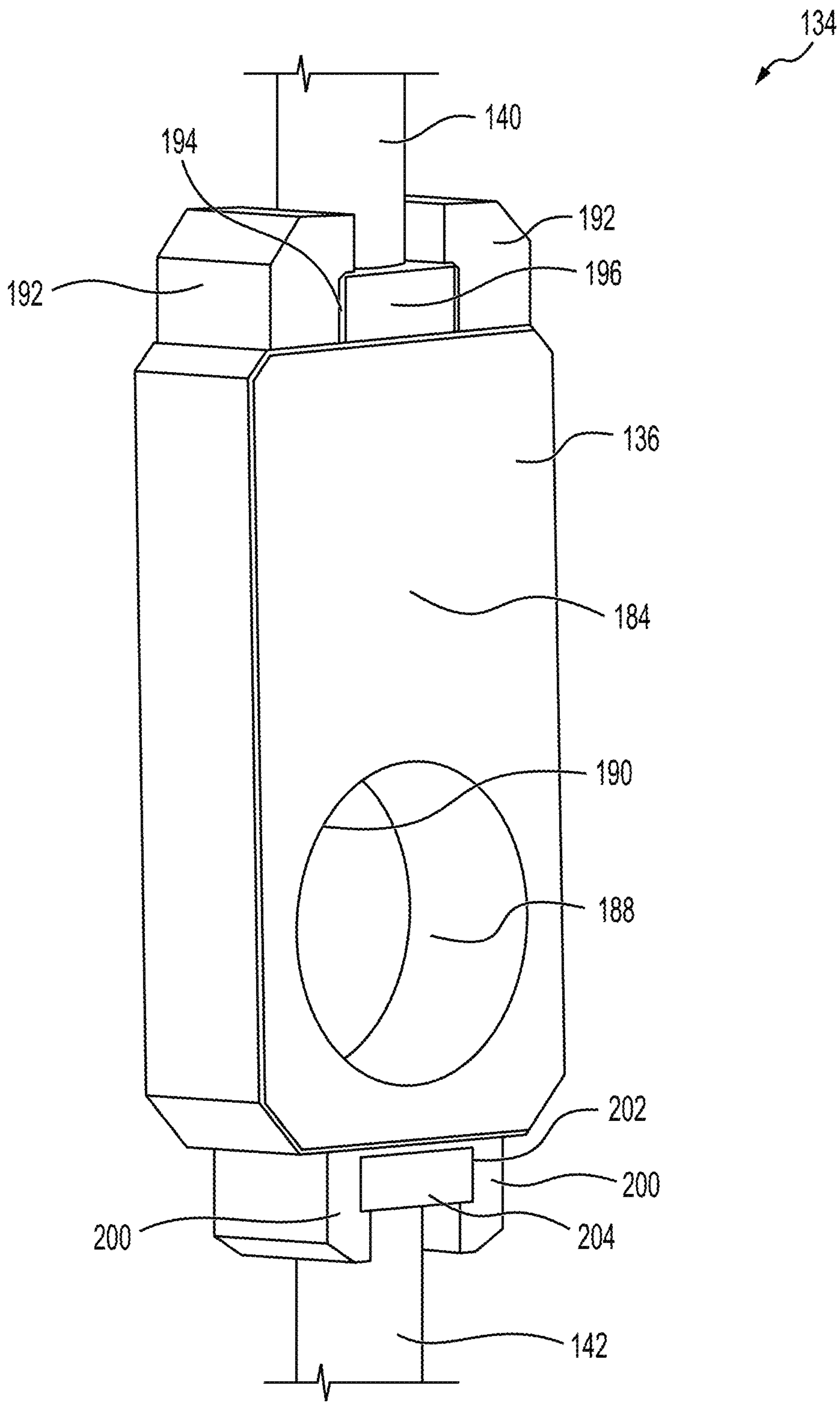
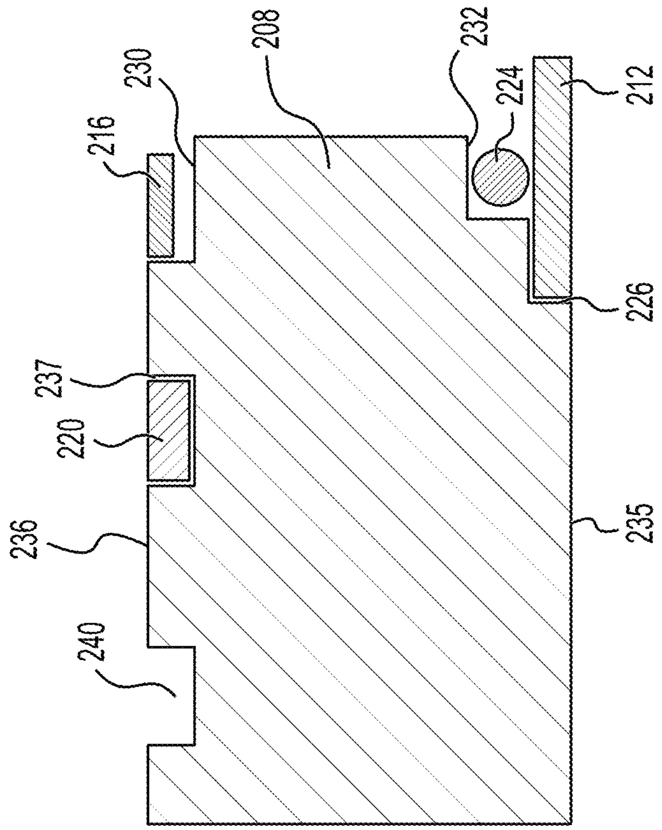


FIG. 5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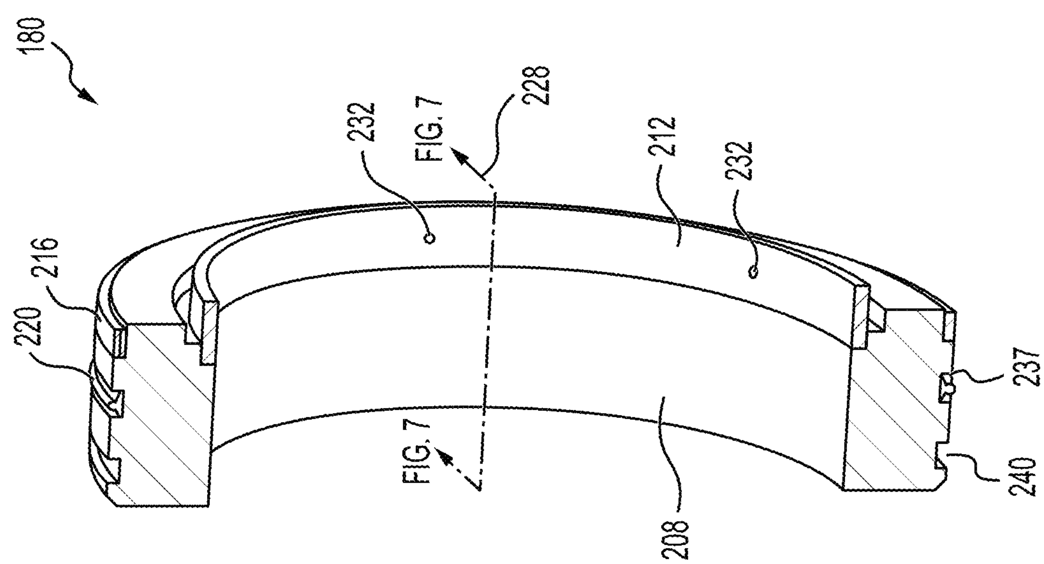
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FIG. 7

FIG. 6

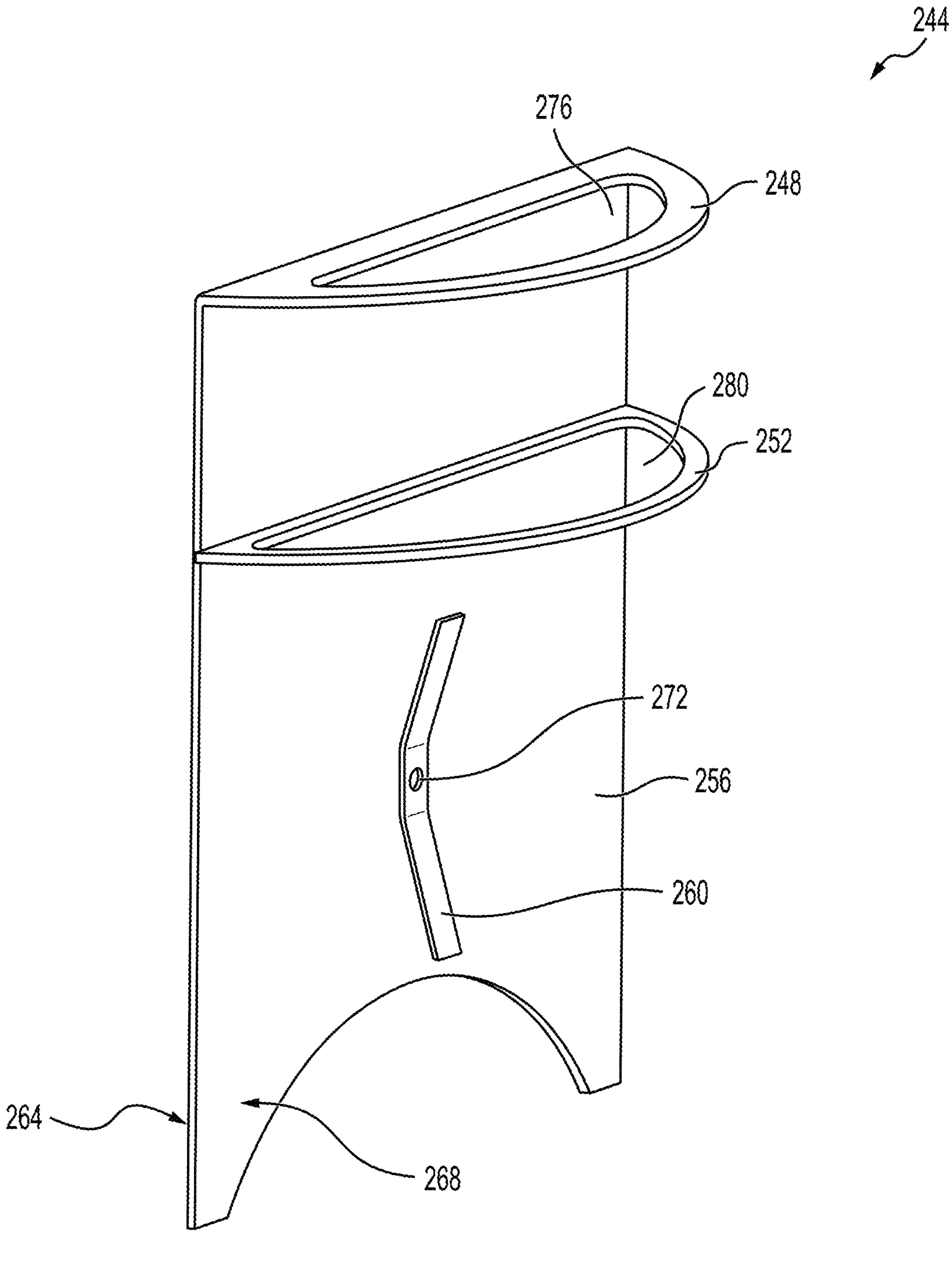

FIG. 8

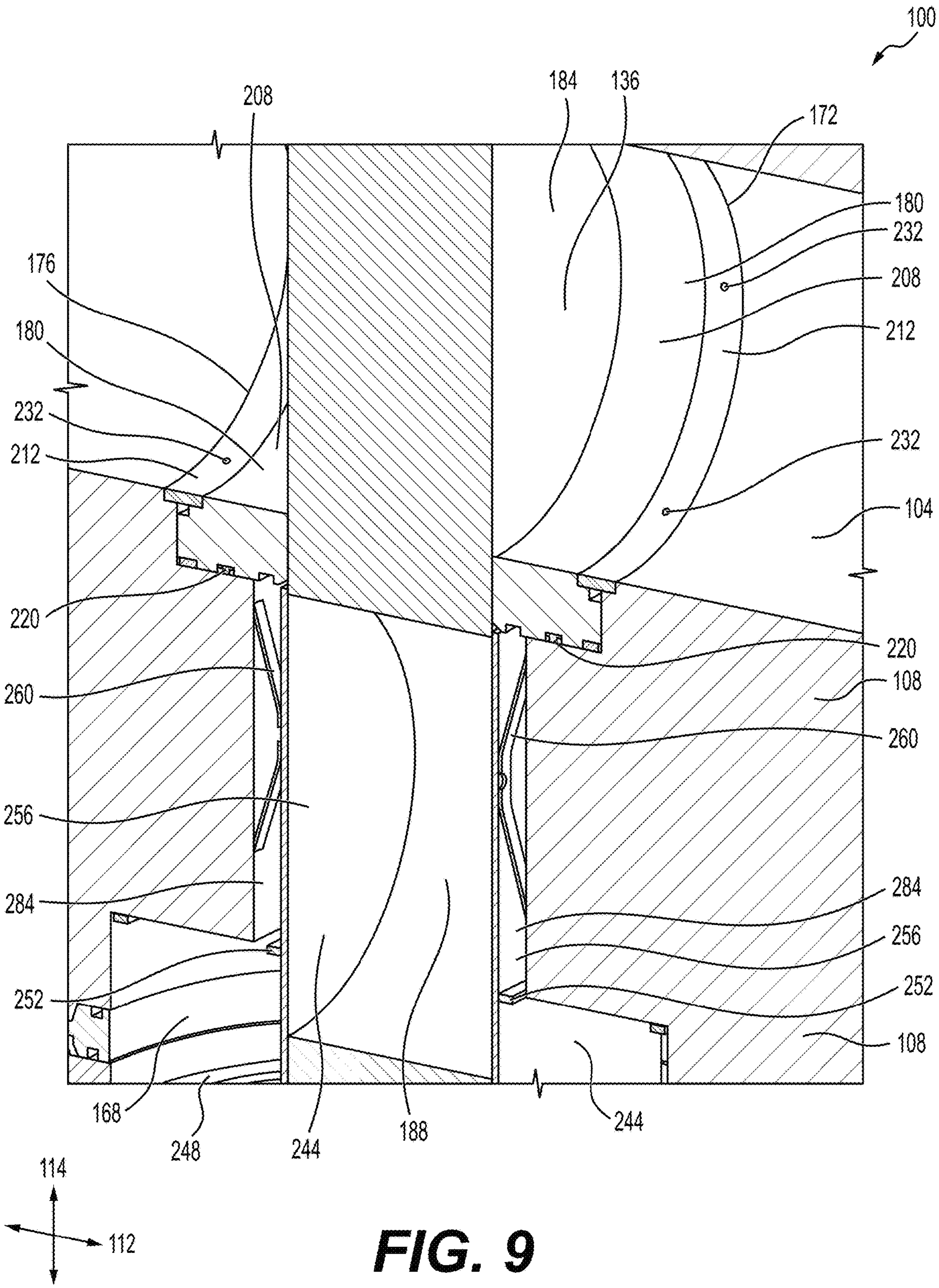

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FIG. 9

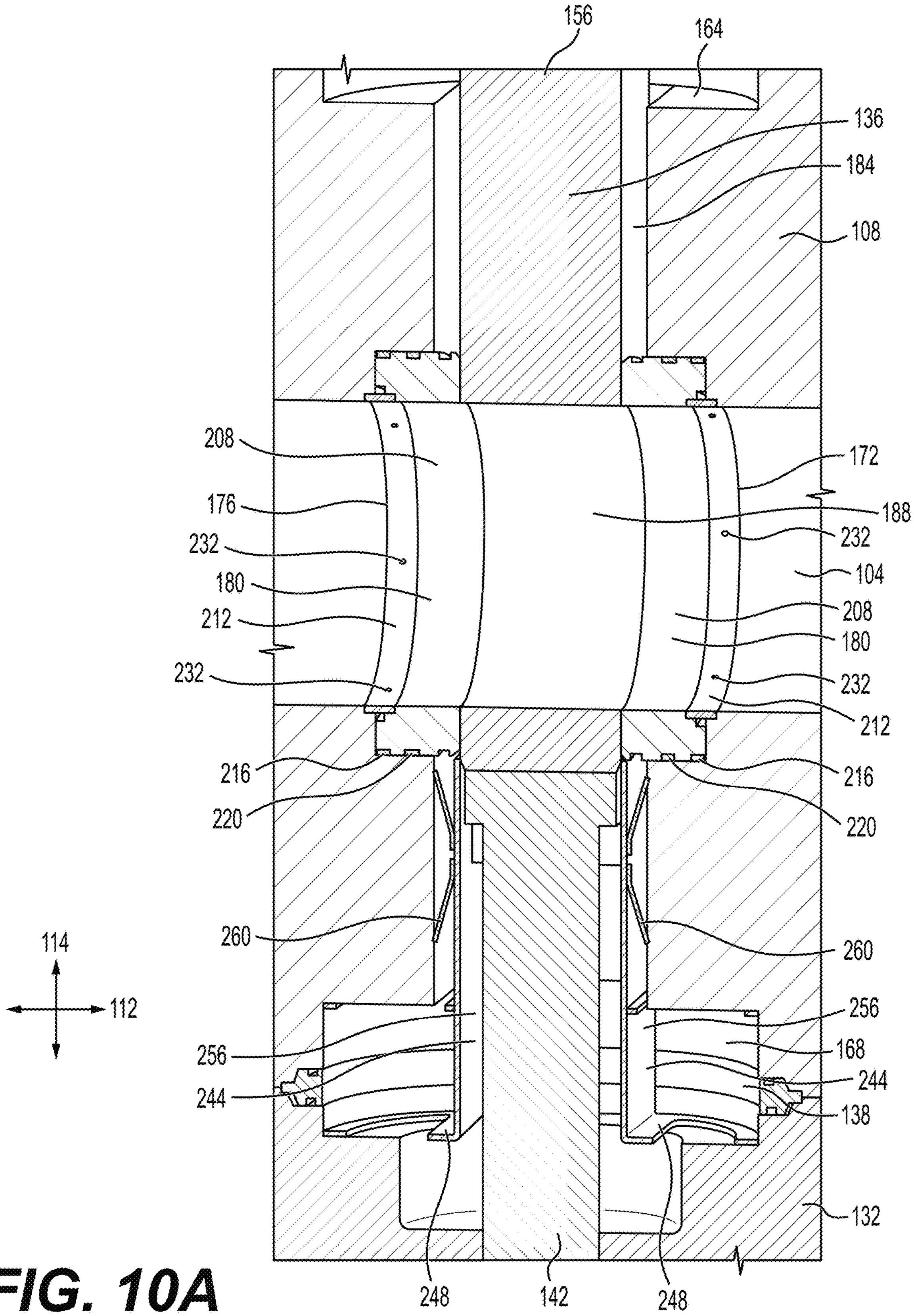
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FIG. 10A

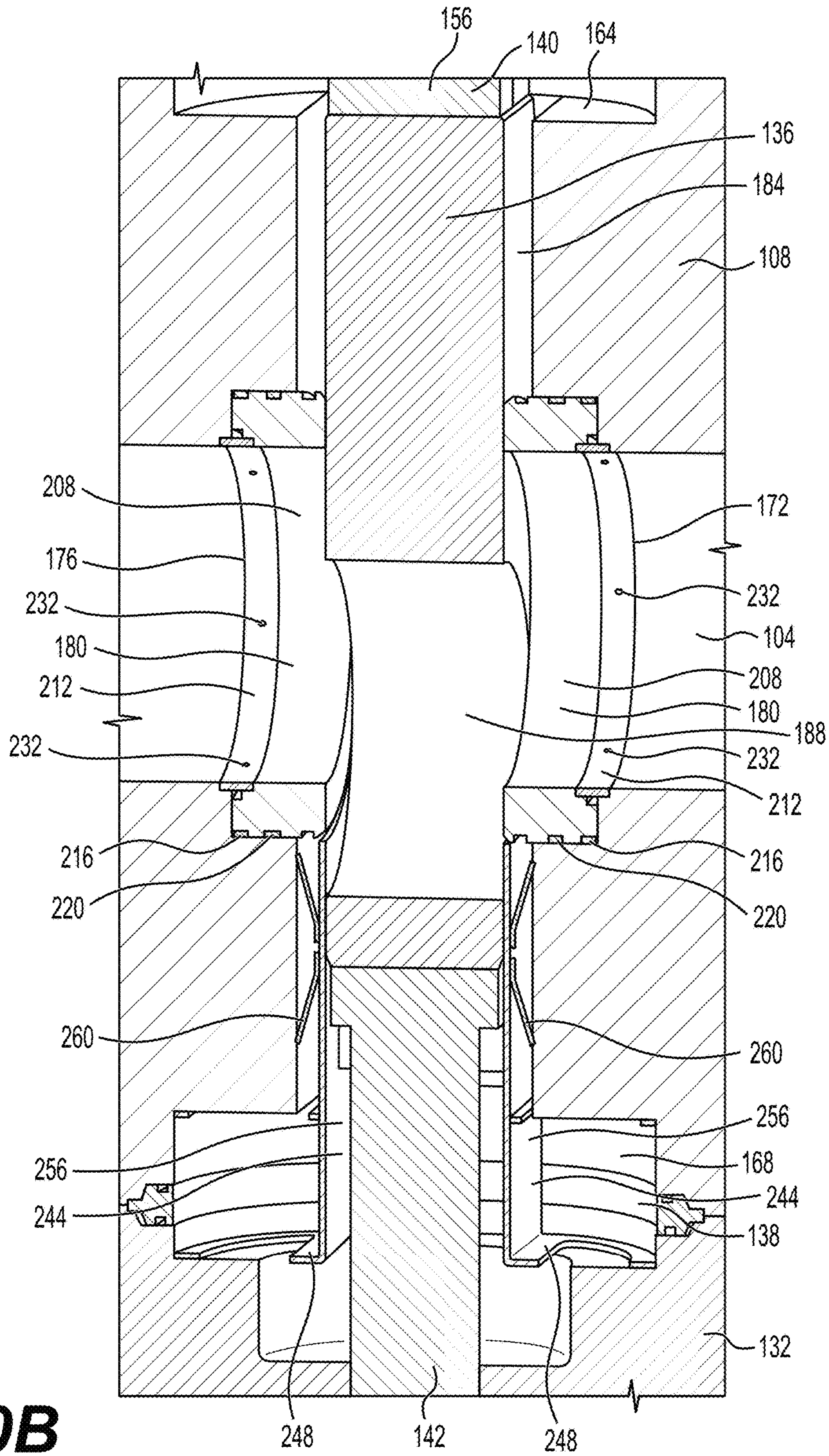
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FIG. 10B

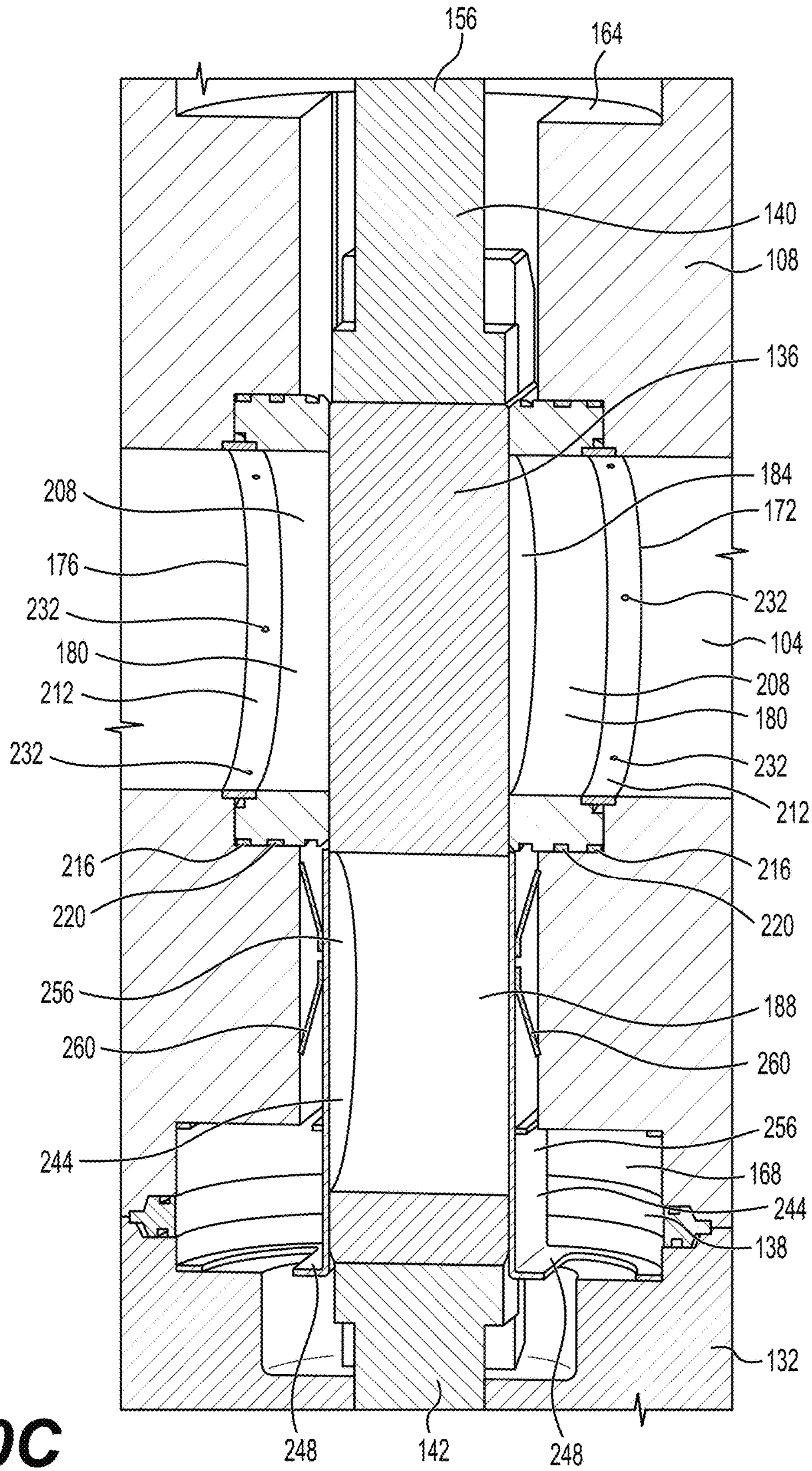
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FIG. 10C

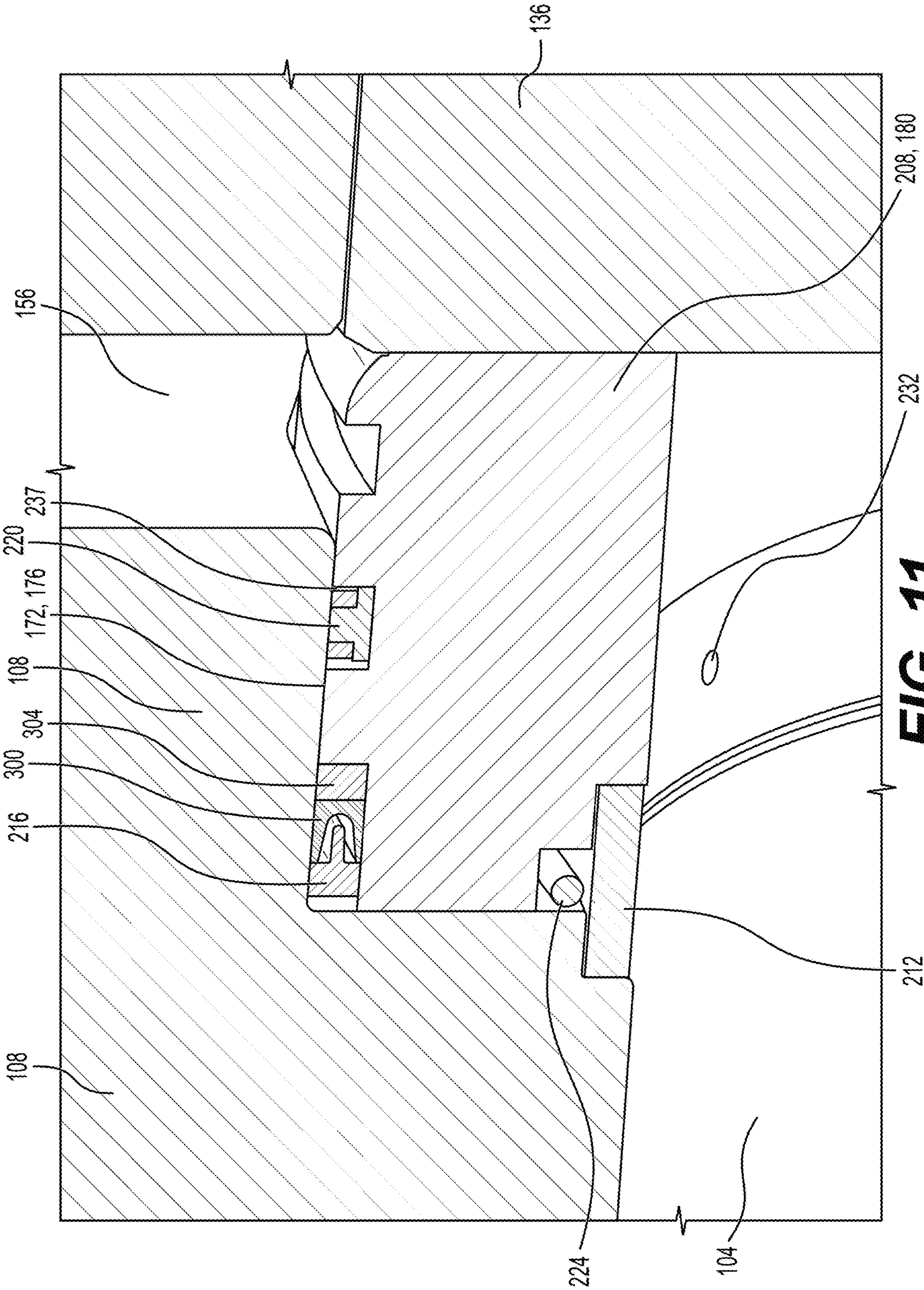
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FIG. 11

DYNAMIC SLAB GATE VALVES

RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 18/416,795 filed Jan. 18, 2024, which claims the benefit of U.S. Provisional Patent Application 63/535,053 filed on Aug. 28, 2023, and U.S. Provisional Patent Application 63/544,777 filed on Oct. 18, 2023, all of which are incorporated herein by reference for all purposes.

TECHNICAL FIELD

This application is directed, in general, to valves for use in fluid transfer applications, and more specifically to dynamic slab gates valves, which may be used in oil and gas applications.

BACKGROUND

The following discussion of the background is intended to facilitate an understanding of the present disclosure only. It should be appreciated that the discussion is not an acknowledgement or admission that any of the material referred to was part of the common general knowledge at the priority date of the application.

Gate valves are used to control the transfer of fluids in tubing and pipelines. Specifically, gate valves are used to stop and start the flow of fluids in a downstream direction. Gate valves are commonly used in the oil and gas industry to control the flow of various fluids such as production fluids, water, fracking fluids, and other fluids used in drilling, operating, and maintaining oil and gas wells.

Gate valves generally operate by actuation of an internal gate, which in one position has an opening to allow upstream fluids to flow through the valve and in a downstream direction and a second position which blocks flow through the valve thereby preventing transfer of fluid in a downstream direction. While gate valves have been in existence for a long time, improvements are still desired.

SUMMARY

According to an illustrative embodiment, a slab gate valve includes a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction, which is orthogonal to the first direction. The slab gate valve further includes a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side. The gate cavity is sized and configured to receive the slab gate. The first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position. The second portion of the slab gate valve is formed with a flow aperture for allowing flow therethrough when the slab gate is in an open position. At least one stem is coupled to the slab gate for selectively moving the slab gate between the open position and the closed position.

The slab gate valve further includes a downstroke lubricant cavity formed in a portion of the valve body; a first seat cavity formed in the valve body proximate a first intersection of the through-bore and the gate cavity on a first side of the first intersection; and a second seat cavity formed in the valve body proximate a second intersection of the through-bore and the gate cavity on a second side of the second intersection. The slab gate valve also includes a first dynamic seat assembly disposed in the first seat cavity, a

second dynamic seat assembly disposed in the second seat cavity, a first dynamic skirt disposed adjacent to the first side of the slab gate and disposed between the first seat cavity and the downstroke lubricant cavity, and a second dynamic skirt disposed adjacent to the second side of the slab gate and disposed between the second seat cavity and the downstroke lubricant cavity.

In one illustrative embodiment, the first dynamic seat assembly in the slab gate valve of the previous two paragraphs may include a first cylindrical seat having a first side and a second side, wherein the second side is adjacent to the first side of the slab gate. A first debris ring slot is formed on an inner portion of the first cylindrical seat. The first dynamic seat assembly may also include a debris ring proximate the first side of the first cylindrical seat and disposed at least partially within the first debris ring slot. The first cylindrical seat may be formed with a seat energizer cavity proximate the first side of the first cylindrical seat. The first dynamic seat assembly may also include a seat energizer disposed at least partially in the energizer cavity and operable to urge the first dynamic seat toward the slab gate.

According to another illustrative embodiment, a slab gate valve includes a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction orthogonal to the first direction and a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side. The gate cavity is sized and configured to receive the slab gate. The first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position. The second portion of the slab gate valve is formed with a flow aperture for allowing flow therethrough when the slab gate is in an open position. The slab gate valve also includes at least one stem coupled to the slab gate for selectively moving the slab gate between the open position and the closed position; a downstroke lubricant cavity formed in a portion of the valve body; at least one seat cavity formed in the valve body proximate an intersection of the through-bore and the gate cavity; and at least one dynamic seat assembly disposed in the at least one seat cavity.

According to still another illustrative embodiment, a slab gate valve includes a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction orthogonal to the first direction and a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side. The gate cavity is sized and configured to receive the slab gate. The first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position. The second portion of the slab gate is formed with a flow aperture for allowing flow therethrough when the slab gate is in an open position. The slab gate valve further includes at least one stem coupled to the slab gate for selectively moving the slab gate between the open position and the closed position; a downstroke lubricant cavity formed in a portion of the valve body; and at least one dynamic skirt disposed adjacent to the slab gate and between the slab gate and the downstroke lubricant cavity. Other valves and embodiments are presented further below.

DESCRIPTION OF THE DRAWINGS

Illustrative embodiments of the present invention are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

FIG. **1** is a schematic perspective view of an illustrative embodiment of a dynamic slab gate valve;

FIG. **2** is a schematic, perspective view with a portion shown in cross section of an illustrative embodiment of a dynamic slab gate valve;

FIG. **3** is a schematic, perspective view with a portion shown in cross section of an illustrative embodiment of a dynamic slab gate valve;

FIG. **4** is a schematic, perspective view with a portion shown in cross section of an illustrative embodiment of a dynamic slab gate valve;

FIG. **5** is a schematic, perspective view of a slab gate and valve stems of an illustrative embodiment of a dynamic slab gate valve;

FIG. **6** is a schematic perspective view with a portion shown in cross section of a dynamic seat assembly of an illustrative embodiment of a dynamic slab gate valve;

FIG. **7** is a schematic cross-sectional view of a dynamic seat assembly of an illustrative embodiment of a dynamic slab gate valve;

FIG. **8** is a schematic perspective view of a dynamic skirt assembly of an illustrative embodiment of a dynamic slab gate valve;

FIG. **9** is a schematic, perspective view with a portion shown in cross section of an illustrative embodiment of a dynamic slab gate valve;

FIG. **10**A is a schematic, cross sectional view of an illustrative embodiment of a dynamic slab gate valve in an open position;

FIG. **10**B is a schematic, cross sectional view of an illustrative embodiment of a dynamic slab gate valve in a half position;

FIG. **10**C is a schematic, cross sectional view of an illustrative embodiment of a dynamic slab gate valve in a closed position; and

FIG. **11** is a schematic, perspective view with a portion shown in cross section of a portion of an illustrative embodiment of a dynamic slab gate valve.

DETAILED DESCRIPTION

In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings that form a part hereof, and in which is shown, by way of illustration, specific embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is understood that other embodiments may be utilized, and that logical structural, mechanical, electrical, and chemical changes may be made without departing from the spirit or scope of the invention. To avoid detail not necessary to enable those skilled in the art to practice the invention, the description may omit certain information known to those skilled in the art. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the claims. Unless otherwise indicated, as used throughout this document, "or" does not require mutual exclusivity.

In one exemplary embodiment, a dynamic gate valve has a valve body with a through-bore running the length of the gate valve. The through-bore being, typically, of circular cross section. The ends of the gate valve are connected to upstream and downstream tubulars, with fluid entering the gate valve from the upstream side and exiting the gate valve from the downstream side.

Flow is opened or closed by a gate. The gate is generally a slab with an opening in one area and a solid surface in another area. The gate resides within a cavity which bisects the through-bore and is able to be moved from an open position, in which the opening of the gate is in line with the through-bore, and a closed position, in which the solid surface of the gate blocks flow through the through-bore.

The gate is a dynamic gate that can translate a distance in the direction of the through-bore in response to fluid pressures exerted against the gate surface. The dynamic gate valve also has a dynamic skirt, which provides a seal between fluid flow areas of the valve and lubricated areas (or lubricated reservoirs or cavities) of the valve. The dynamic skirt, therefore, reduces cross-flow between through-bore fluids and lubricants, thereby reducing contamination of lubricants and loss of lubricants. The dynamic skirt is biased by a skirt energizer, so that the dynamic skirt is biased toward the gate to achieve a seal with a gate surface and to move in response to gate movement.

The dynamic gate valve may also include a dynamic seat, which provides a seal between the gate and the through-bore to prevent through bore fluid from flowing into operational areas of the valve. Part of the seal formed by the seat includes contact of a seat surface with the gate, which prevents through-bore fluid from flowing into lubrication areas. The seat is biased by an energizer toward the gate surfaces to increase the quality of this seal. In addition, the biasing force of the energizer results in movement of the seat toward the gate as the gate translates in the direction of the through-bore. Thereby, the seat dynamically moves with the movement of the gate, which moves in response to fluid pressures, to maintain the seal between the seat and the gate.

Referring now to the figures and primarily to FIGS. **1-4**, an illustrative embodiment of a slab gate valve **100** is presented. The slab gate valve **100** has a through-bore **104**, which is a circular bore through the length of a valve body **108**, for allowing a fluid flow therethrough as desired. The through-bore **104** is oriented along the length of the valve body **108** in a first direction **112**. The slab gate valve **100** is coupled to a pipe or other tubing, through which fluid may flow in the first direction **112** with a first flange **116** and a second flange **120**. A pipe or other tubing is connected to the first flange **116** and the second flange **120** with bolts or studs using a plurality of bolt holes **124**, so that fluid may flow through the through-bore **104** in the first direction **112**. The valve body may be connected to a pipe via any attachment technique or device including, inter alia, the following: Welded Connection, Studded Connection, Hammer Union, Threaded Connection, Hub & Clamp Connection, and Collar Connection. The disclosure also includes a multi valve body comprising of one or more valves in a singular block body.

It should be noted that slab gate valve **100** may be bi-directional, but for convenience, is presented with flow in a single direction. A gasket pocket **152** is disposed within the end of each flange **116** and **120** so that a gasket may be inserted into the gasket pocket **152** to provide a fluid seal between the first flange **116** or the second flange **120** and an upstream or downstream pipe or tubing connection, e.g., a mating flange.

An upper bonnet **128** and a lower bonnet **132** are located on an upper side and a lower side, respectively, of the valve body **108**. All relative directions, e.g., "upper" and "lower," are for the orientation shown in the figures. The upper bonnet **128** and the lower bonnet **132** are used to attach and contain a portion of a gate assembly **134** within the valve body **108**. The gate assembly **134** is formed of a gate **136**, an upper stem **140**, and a lower stem **142**. The upper bonnet **128** and the lower bonnet **132** are coupled to the valve body

108 with studs or bolts using a plurality of bolt holes **144**. The upper stem **140** and the lower stem **142** extend from the centers of the upper and lower ends of the upper bonnet **128** and the lower bonnet **132**, respectively. The upper stem **140** and the lower stem **142** are used to actuate the gate assembly **134** between an open and closed position.

A gate cavity **156** is a cavity within the valve body **108** that is oriented in a second direction **114** and that is perpendicular to and transects the through-bore **104**. The gate cavity **156** is sized and configured to receive a gate assembly **134**, and the gate cavity **156** may be further sized and configured to receive a dynamic shield assembly and may be optimized to minimize or reduce the volume of lubricant required. An upstroke lubricant cavity **164** and a downstroke lubricant cavity **168** are partially formed by cutouts from the valve body **108** at the upper and lower ends of the gate cavity **156**. The upstroke lubricant cavity **164** and the downstroke lubricant cavity **168** are also partially formed by cutouts within the upper bonnet **128** and the lower bonnet **132**, respectively. When the upper bonnet **128** and the lower bonnet **132** are assembled onto the valve body **108**, the cutouts within the upper bonnet **128** and the valve body **108** and the cutouts within the lower bonnet **128** and the valve body **108** align to form the upstroke lubricant cavity **164** and the downstroke lubricant cavity **168**, respectively. The connection between the upper bonnet **128** and the valve body **108** is sealed from fluid flow by an upper bonnet gasket **130**. Likewise, the connection between the lower bonnet **132** and the valve body **108** is sealed from fluid flow by a lower bonnet gasket **138**. The lower bonnet gasket **138** and the upper bonnet gasket **130** are made from suitable gasket material, such as any compressible material, for example any of the NBR type rubbers. Other examples include metallic, non-elastomeric materials (PTFE, MPTFE, PEEK, Graphite/re-enforced Graphite, or any combination of the above. Those skilled in the art will appreciate that other materials may be used as well.

The through-bore has a diameter **148**, which is of sufficient size to allow for the desired flow through slab gate valve **100**. In one embodiment the diameter **148** is in the range of 1¹³⁄₁₆" thru 7¹⁄₁₆". Those skilled in the art will appreciate that other sizes may be used.

The upstream seat pocket **172** and the downstream seat pocket **176** are cavities within the valve body **108** within the through-bore **104** at the intersection of the through-bore **104** and the gate cavity **156**. The upstream seat pocket **172** and the downstream seat pocket **176** are both configured to receive a dynamic seat assembly **180**, which provides fluid and pressure seals between the seat assemblies **180** and the gate assembly **134**. The upstream seat pocket **172** and the downstream seat pocket **176** are annular rings that are concentric with the through-bore **104**.

As the slab gate valve **100** is operated between an open and a closed position, the gate assembly **134** moves back and forth along the second direction **114** within the gate cavity **156**. This movement results in movement of the upper end of the gate **156** and the upper stem **140** through upstroke lubrication cavity **164** and movement of the lower end of gate **156** and the lower stem **142** through downstroke lubrication cavity **168**. In the course of this movement, the gate **156**, the upper stem **140**, and the lower stem **142** are lubricated by lubricants within upstroke lubrication cavity **164** and downstroke lubrication cavity **168**. A dynamic skirt assembly **244**, which is discussed in more detail below, acts as a barrier to prevent excessive exchange of lubricants and fluids between the through-bore **104** and the downstroke lubrication cavity **168** and the upstroke lubrication cavity **164** during the opening and closing process.

Referring now primarily to FIG. **5**, an illustrative embodiment of the gate assembly **134** is presented. The gate assembly **134** contains the gate **136**, the upper stem **140**, and the lower stem **142**. The gate **136** has a closed portion **184** and an open portion **188**. The open portion **188** contains a circular cutout or flow aperture **190** to allow for flow through the gate **136**. When the gate **136** is in the open position, the open portion **188** is in line with and concentric to the through-bore **104**. The closed portion **184** is solid to prevent flow through the gate **136**. When the gate **136** is installed into the slab gate valve, such as slab gate valve **100**, and is in the closed position, the closed portion **184** of gate **136** is in line with and blocks flow though the through-bore **104** of the slab gate valve **100**. When gate **136** is installed into a valve, such as slab gate valve **100**, and is in the open position, the open portion **188** of gate **136** is in line with and allows for flow though the through-bore of the slab gate valve **100**. In the depicted embodiment, the closed portion **184** is depicted as being above the open portion **188**. In other embodiments, this relationship between the closed portion **184** and the open portion **188** can be reversed so that closed portion **184** is below open portion **188**.

An upper end of the gate **136** is configured to receive the upper stem **140**. Shoulders **192** may be formed on the upper end of the gate **136**. The shoulders **192** are configured to form a t-slot for receiving a portion of the upper stem **140**, namely a tee **196**. The lower end of the upper stem **140** has the tee **196**. The upper stem **140** is coupled to and captured by the gate **136** by sliding the tee **196** of the upper stem **140** into the tee slot **194** formed by the shoulders **192**. This configuration allows for the gate **136** to be captured by the upper stem **140** so that the movement of the upper stem **140** within slab gate valve **100** in the first direction **112** of the gate cavity **156** causes the gate **136** to move along with the upper stem **140**, therefore allowing upper stem **140** to be used to move the gate **136** between a closed and open position within the slab gate valve **100**.

The lower end of the gate **136** is similarly configured with shoulders **200** forming a tee slot **202** for receiving a tee **204**. In this way the gate **136** receives the lower stem **142**, so that the lower stem **142** may also be used to move the gate **136** between and open and closed position within a valve.

While, in this embodiment the tee and t-slot connection between gate **136** and upper stem **140** and gate **136** and lower stem **142** is formed with the tee portion on the upper stem **140** and on the lower stem **142** and the t-slot portions on the gate **136**, in other embodiments this relationship can be reversed so that the tee portions are formed on the gate **136** and the t-slot portions are formed on the upper stem **140** and the lower stem **142**. In other embodiments, the gate assembly **134** has only one stem connection, which may be either an upper stem or lower stem.

The tee **196** and the tee **204** allow for movement in the first direction **112** (flow direction) of the gate **136** relative to the upper stem **140** and the lower stem **142** to accommodate for pressures applied to the gate **136** when the gate **136** is in use in a valve, such as slab gate valve **100**.

Referring now primarily to FIGS. **6** and **7**, an illustrative embodiment of the dynamic seat assembly **180** of a slab gate valve **100** is presented. FIG. **6** depicts a perspective view with a portion in cross section of the dynamic seat assembly **180**. FIG. **7** depicts a cross sectional view of the dynamic seat assembly **180** taken at the cross-section location **228** (FIG. **6**). When installed within the slab gate valve **100**, the dynamic seat assemblies **180** are disposed within an upstream seat pocket **172** and a downstream seat pocket **176**.

The embodiment of the dynamic seat assembly **180** depicted in FIGS. **6** and **7** has a seat **208**, a first debris ring **212**, a second debris ring **216**, a seat seal **220**, and a seat energizer **224**. The seat **208** fits with a seal pocket (sec **172** and **176** in FIG. **4**). The first debris ring **212** may be disposed at least partially in the first debris ring slot or shoulder **226**. The second debris ring **216** may be disposed at least partially within a second debris ring shoulder or slot **230**. The seat energizer **224** may be disposed at least partially in a seat energizer slot of the shoulder **232**, which also may be referred to as a seat energizer cavity, and may be adjacent to a portion of the first debris ring **212**. The seat **208** has a cylindrical body with an inner through-bore. The seat **208** has an inner portion **235** and outer portion **236**, or perimeter.

The first debris ring **212** and the second debris ring **216** each provide a debris seal between the valve body **108** and the seat **208**. The first debris ring **212** is perforated by the perforations **232** to allow pressure through the first debris ring **212**, while preventing large particles from passing through the seal. The perforations may be in the range of 063 inches or smaller and may be formed by machining via drill, waterjet, laser cutter, or other technique. The second debris ring **216** may have no perforations and may be sized to maintain a space between the seat **208** outside diameter and the valve body seat pocket **176** ID. The space is designed to prevent even smaller material from entering the seal area **220**/**237**. The gap may be in the range of 0.003" or smaller.

In addition, fluid may flow past first debris ring **212** and the second debris ring **216** in the space between first debris ring **212** and the second debris ring **216** and the valve body **108** or seat **208**. The first debris ring **212** or the second debris ring **216** can serve a dual role as a bearing and as a device to prevent debris from reaching seat seal **220**. In some embodiments, the dynamic seat assembly **180** may have only one debris ring. In some embodiments, the first debris ring **212** and the second debris ring **216** are not perforated.

The first debris ring **212** and second debris ring **216** are configured to exclude the passage particles and debris of a certain size past the first debris ring **212** or the second debris ring **216**. The minimum size of the particles excluded by the first debris ring **212** is larger than that excluded by the second debris ring **216**. Therefore, the first debris ring **212** acts as a first pass particulate filter and the second debris ring **216** acts as a second pass particulate filter, which removes finer particles than are removed by the first debris ring **212**. In some embodiments the first debris ring **212** prevents debris in fluids that are flowing through the through-bore **104** that are 0.063" in diameter or larger from passing the first debris ring **212**. In some embodiments, the second debris ring **216** prevents debris in fluids that are flowing through the through-bore **104** that are 0.002"-0.003" in diameter from passing the second debris ring. Those skilled in the art will appreciate that other sizes of apertures may be used.

The first debris ring **212** and second debris ring **216** may be made from many materials, e.g., PEEK (polyetheretherketone) material (or other suitable plastic) or metals. In one embodiment, the debris rings **212**, **216** are made from graphite or re-enforced graphite.

The connection between the dynamic seat assembly **180** and the valve body **108** is scaled, at least in part, from fluid and gas flow by the seat seal **220**. The seat seal **220** is disposed within a seat seal cavity **237**, which is formed in an exterior or outer perimeter of the seat **208**. When the dynamic seat assembly **180** is installed within the upstream seat pocket **172** (FIG. **4**) or the downstream seat pocket **176**, the seat seal forms a liquid and gas tight seal to prevent or at least substantially restrict fluid and gas flow.

The seat seal **220** can be either non-elastomeric material like polytetrafluoroethylene (PTFE), modified polytetrafluoroethylene (MPTFE), polyetheretherketone (PEEK), graphite/re-enforced graphite, or metal or a combination or made of an elastomer like that used for O-rings, e.g., hydrogenated nitrile butadiene rubber (HNBR), nitrile butadiene rubber (NBR), fluo Kohlenstoff material (FKM), perfluorelastomers (FFKM), etc. The seat seal **220**, the first debris ring **212**, the and second debris ring **216** can vary with different applications. In one embodiment, the seat seal **220** is about ¼ inch in width, 9 inches in diameter, and ⅛ deep.

The dynamic seat assembly **180** also may include a tooling cutout **240**. The tooling cutout **240** allows for a tool to be inserted into the tooling cutout **240** for prying the dynamic seat assembly **180** from the upstream seat pocket **172** or the downstream seat pocket **176**.

Referring now primarily to FIG. **8**, an illustrative embodiment of a dynamic skirt assembly **244** of a dynamic slab gate valve, such as slab gate valve **100** is presented. The dynamic skirt assembly **244** may include a first lubricant guide **248**, and may include a second lubricant guide **252**, a skirt plate **256**, and a skirt energizer **260**. The skirt plate **256** has a first side **264** that when assembled is closest to the gate **136** (FIG. **3**) and has a second side **268** that when assembled is furthest from the gate **136**, when installed into slab gate valve **100**. The first side **264** of the skirt plate **256**, when installed in slab gate valve **100**, faces the gate **136** within the valve to provide a sealed surface between the skirt plate **256** and the gate **136** to prevent excessive grease and fluid flow between such seal. The second side **268** of the skirt plate **256** faces a portion of the valve body **104** (see FIG. **3**). The skirt energizer **260** is coupled to the skirt plate **256** on the second side **268**. In the depicted embodiment, this coupling may be made by a spot weld **272**. Other coupling methods, such as screws, weld-stud, or a nut and bolt or other fasteners may be used for this coupling. The skirt energizer **260** may be coupled vertically (for orientation shown) as shown in FIG. **8** or horizontally (i.e., a third direction **115** in FIG. **3**). The first lubricant guide **248** and the second lubricant guide **252** extend from the second side **268** of the skirt plate **256** and are shaped to conform with the shape of lubricant pockets of a valve, such as downstroke lubricant cavity **168** of FIG. **2**. The first lubricant guide **248** and the second lubricant guide **252** have openings **276** and **280**, respectively to allow lubrication, such as grease to flow through the openings **276** and **280**. The lubricant guide **248**, **252** may be welded, bolted, mechanically attached, or formed directly to the skirt plate **256**. A lower portion of the skirt plate **256** may be formed with or without a cutout **258**, e.g., an arcuate cutout as shown.

FIGS. **9**, **10**A, **10**B, and **10**C depict the operation of a representative embodiment of a dynamic skirt assembly **244** of a slab gate valve **100**. As depicted in the proceeding figures, the dynamic skirt assembly **244** is located within the slab gate valve **100** and in particular is partially disposed between the gate **136** and a valve body **108** and partially disposed between a lower stem **142** and the valve body **108**. A portion of the dynamic skirt assembly **244** is also disposed within the downstroke lubricant cavity **168**, the downstroke lubricant cavity **168** being formed from cavities in the valve body **108** and in a lower bonnet **132** and, when the slab gate valve **100** is in use, being filled with a lubricant, such as grease. The dynamic skirt assembly **244** is comprised of the skirt plate **256** and a skirt energizer **260**, with the side of the dynamic skirt assembly **244** having the skirt energizer **260** facing the valve body **108** so that the skirt energizer **260** is in contact with the valve body **108** and the skirt energizer **260** biases the dynamic skirt assembly **244** away from the valve body **108**, therefore, creating cavity **284** between the skirt plate **256** and the valve body **108**. The force of the skirt energizer **260** also biases the other side of the skirt plate **256** toward the gate **136** and the lower stem **142**, thereby creating a seal between the skirt plate **256** and the gate **136** or the lower stem **142**. In one illustrative embodiment the skirt energizer **260** is a metal spring formed with angled ends that have a free position lifted off the skirt plate **256**. Other types of energizers may be used or wire-type (coiled or bent) compression, conical, leaf, Belleville, grater, disc type energizers that may be made from rubber, plastic, metal, or another material.

As depicted in FIG. **9**, fluid pressure in the through-bore **104** creates a pressure force against the upstream side of the gate **136**. As described above in relation to FIG. **5**, the connection between the gate **136** and lower stem **142** and the gate **136** and the upper stem **140** is a dynamic connection where the gate **136** can translate in the first direction **112** along the length of the through-bore **104** in response to pressure forces from fluids within the through-bore **104**. Therefore, when the gate **136** is in the closed position, upstream fluid pressure pushes against the gate **136**, which results in translation of the gate **136** in the first direction **112**, as the tee slides relative to the associated slot (see FIG. **5**).

The dynamic skirt assembly **244** is intended to provide a good seal between the fluid and debris flowing through the through-bore **104** and the lubricant located in the downstroke lubricant cavity **168**. A poor seal between the gate **136** and the skirt plate **256** or between the lower stem **142** and the skirt plate **256** results in the entry of through-bore fluids and debris into the downstroke lubricant cavity **168** and for the loss of lubricant into the through-bore **104**, both of which result in decreased efficiency and operation of the slab gate valve **100**. The potential for such to happen is particularly an issue during opening and closing operations of the slab gate valve **100**.

Referring now primarily to FIGS. **10**A, **10**B, and **10**C, an illustrative embodiment of a slab gate valve in various positions (an open position, a partially open position, and a closed position, respectively) is presented. When the gate **136** is in the open position, fluid is flowing through the through-bore **104** and the open portion **188** of the gate **136**. As the gate **136** is moved toward a closed position (see, e.g., the intermediate position in FIG. **10**B), the open portion **188** of the gate **136** moves lower into the gate cavity **156** and eventually, once the closed position (see, e.g., FIG. **10**C) is reached, is totally located below the through-bore **104** and is not in fluid communication with the through-bore **104**. However, during the closing process, fluid that is located within the open portion **188** of the gate **136** is trapped within the opening **188** and remains trapped within the open portion **188** when the gate **136** is in the closed position.

Without the skirt plate **256** in place as depicted in FIG. **10**C, the trapped fluid in the open portion **188** can commingle and mix with the lubricant located in the downstroke lubricant cavity **168**. This results in the introduction of through-bore fluids and debris into the downstroke lubricant cavity **168**, which contaminates and reduces the effectiveness of the lubricant. In addition, this results in the dilution and washout of lubricant from the downstroke lubricant cavity **168**, with lubricant moving into the open portion **188**, where the lubricant serves no effective purpose.

The existence of the skirt plate **256** assists in reducing the introduction of through-bore fluids and debris into the lubricant and the loss of lubricant from the lubricant pocket, or cavity. The skirt plate **256** provides a physical barrier to reduce or prevent this exchange.

However, to prevent this exchange and loss, a reliable seal between the skirt plate **256** and the gate **136** or the lower stem **142** is needed. Also, as discussed above, the gate **136** is a dynamic gate that moves in the first direction **112** in response to through-bore **104** fluid pressure. In this situation, a static skirt plate is inadequate. In the case of a static skirt plate, the skirt plate and a gate that make contact when fluid pressure is not applied to the gate do not make sufficient contact when fluid pressure causes the gate to move in the downstream direction.

The dynamic skirt assembly **244** addresses this deficiency. The skirt energizer **260** provides a biasing force between the valve body **104** and the skirt plate **256**. This biasing force results in a tighter, closer seal with the gate **136**. Furthermore, as the gate **136** is moved in the first direction **112** by fluid pressures, the biasing force of the skirt energizer **260** causes the skirt plate **256** to move in conjunction with the gate **136**, which in turn results in maintaining good contact and seal between the gate **136** and the skirt plate **256**. This, in turn, reduces the amount of through-bore fluid that enters the downstroke lubrication cavity **168** and the amount of lubricant that is lost into the open portion **188** of the gate **136**.

While in the depicted embodiments of the slab gate valve **100**, the open portion **188** of the gate **136** is above the closed portion **184** of the gate **136**, and the gate **136** is moved from an open position to a closed position by moving the gate **136** in the second direction **114** towards the downstroke lubricant cavity **168**. The direction of movement and the location of the open portion **188** relative to the closed portion **184**, as described, is for reference in relation to the embodiments as depicted. For example, a reversal of the relative positioning of the open portion **188** to the closed portion **184** and a reversal of the movement of the gate **136** from the open position to the closed position, would describe the embodiments depicted in FIGS. **2** and **3**, if the embodiments of FIGS. **2** and **3** were inverted.

Referring now again primarily to FIG. **9**, which also depicts the function of the dynamic seat assembly **180**, the dynamic seat assemblies **180** are disposed within upstream seat pocket **172** and downstream seat pocket **176** of the slab gate valve **100**. As discussed above, the connection between the upper stem **140** and the gate **136** and the connection between the lower stem **142** and the gate **136** is a tee and t-slot type connection, which allows for the gate **136** to move in first direction **112** in response to fluid pressure of fluids within through-bore **104**. Contact between the gate **136** and the side of the dynamic seat assembly **180** that is adjacent to the gate **136** provides a seal to prevent through-bore fluids from transferring into the portions of the gate cavity that are above and below the level of the through-bore **104** within the valve body **108**, such as the upstroke lubricant cavity **164** and the downstroke lubricant cavity **168**. As discussed above fluid communication between the upstroke lubricant cavity **164** and the through-bore **104** or between the downstroke lubricant cavity **168** and the through-bore **104** results in loss of lubricant from the downstroke lubricant cavity **168** or the upstroke lubricant cavity **164** and in contamination of the fluids within through-bore **104**.

A static seat assembly, may provide such a seal when fluid pressure is not applied to a gate of valve. However, as the gate moves in response to fluid pressures, the seal between the gate and a static seat assembly is lost because the movement of the gate results in a gap between the gate and the seat assembly.

The use of the dynamic seat assembly **180** addresses this issue. The dynamic seat assembly **180** includes a seat energizer **224**, which, when installed in the slab gate valve **100**, is disposed between the seat **208** and the valve body **108**. The seat energizer **224** provides a biasing force that urges the seat **208** towards the gate **136**. The seat energizer may be any suitable energizer capable of providing a biasing force between the valve body **108** and the seat **208**. In some embodiments the energizer is a wave spring. Other types of energizers may be wire type (coiled or bent) compression, conical leaf, Belleville, grater, disc-type, which may all be made from rubber, plastic, material or another material.

This biasing force from the seat energizer may result in better seal quality between the seat **208** and the gate **136**. In addition, as the gate **136** is pushed in a downstream direction, the biasing force of the seat energizer **224** causes the seat **208** to move in conjunction with the gate **136** and, therefore, maintain a seal between the seat **208** and the gate **136**.

As used herein, "seal" means a substantial restriction of fluids or other substances; for example, in one example, a seal means less than 0.5% of fluid (gaseous, solid, or liquid) flowing gets by the seal.

Referring now primarily to FIG. **11**, a portion of an illustrative embodiment of a dynamic slab gate valve is shown. In the depicted embodiment, an alternative scaling configuration is depicted regarding the seal between the valve body **108** and the seat **208**. Like in the previously described embodiments, the seat assembly **180** is disposed within a seat pocket, which may be an upstream seat pocket **172** or downstream seat pocket **176**. The seat assembly **180** has a seat **208**, a first debris ring **212**, and a seat seal **220**. The seat seal **220** is disposed within the seat seal cavity **237**. The seat seal **220** is formed from an elastomeric material that forms a seal between the seat **208** and the valve body **108**. The contact seals between the seat seal **220**, the seat **208**, and the valve body **108** act as a liquid seal, a gas seal, and a pressure seal. The first debris ring **212** may have perforations **232**, which function as described above in relation to other embodiments to prevent the travel of debris from fluid flowing in the through-bore **104**. The seat assembly **180** also has a seat energizer **224**, which functions as described above to bias the seat **208** toward the gate **136**.

The seat assembly **180** also has a second debris ring **216**, a u-seal **300**, and a u-seal backer **304**. The u-seal **300** is formed from a material that does not allow gas permeation through the u-seal **300**. In some embodiments, the u-seal **300** is non-elastomeric. In some embodiments, the u-seal **300** is made from PTFE or PEEK. The u-seal **300** is formed so that a cross section of the u-seal **300** has a u-shaped profile, so that one side of the u-seal forms a cup or cavity. The u-seal **300** is installed within the seat assembly **108** so that the cup or cavity side of the u-seal faces the direction of fluid flow from the first debris ring **212** toward the second debris ring **216**. As fluid flows past the second debris ring **216** and to the u-seal **300** (left to right for the orientation shown), the cup or cavity portion of u-seal **300** is filled with fluids, which are under pressure. The fluid pressure is then exerted on the inside wall of the u-shaped seal **300**, which in turn causes the walls of the u-seal **300** to expand toward the valve body **108** and the seat **208**. The force of the fluid pressure pushing the walls of the u-seal **300** against the valve body **108** and the seat **208** results in a liquid and gas tight seal between the u-seal **300** and the valve body **108** and the u-seal **300** and the seat **208**. The higher the pressure that is exerted on the u-seal **300**, the tighter the seal becomes.

In the depicted illustrative embodiment, the second debris ring **216** is formed with a prong that is designed to fit within and conform to the cup or cavity shape of the u-seal **300**. In these embodiments, the prong of the second debris ring **216** prevents undesired deformation of the u-seal **300** when pressure is applied to the u-seal **300** or directs the proper expansion of the walls of the u-seal **300** when fluid pressure is applied within the cup or cavity portion of the u-seal **300**. In some embodiments, the second debris ring **216** is able to slide in response to fluid pressure so that the prong portion of the second debris ring **216** is forced into the cup or cavity portion of the u-seal **300** and makes contact with the u-seal to promote the proper expansion of the walls of the u-seal when pressure is applied, as described above.

The illustrative embodiment of FIG. **11** may provide certain advantages. In particular, it is often necessary to form the seat seal **220** from an elastomeric material. This ensures that seat seal **220** can flex and conform to maintain a liquid and gas seal between the seat seal **220** and the valve body **108** and between the seat seal **220** and the seat **180**. However, the use of elastomeric seals in oil and gas operations potentially suffers from a fault. In particular, elastomeric materials that are suitable for forming the gas and liquid seals described are vulnerable to gas permeation. As gas pressure is applied to the elastomeric material, the gas is forced into and permeates through the elastomeric material. In applications in the oil and gas industry, the source of such gas can be gasses flowing through a pipeline or can be dissolved gasses present in fluids that are flowing through a pipeline.

In either case, since the seat seal **220** is often formed of an elastomeric material, while the seat seal **220** provides a reliable contact seal to prevent liquid and gas flow, it is subject to the possibility of gas flow through the seal **220** by permeation and release with pressure changes. This results in gas flow past the seat seal **220** and into gate cavity **156**, which is not desired.

On the other hand, many non-elastomeric materials, such as PTFE, PEEK, and other plastics, are not subject to gas permeation. By making u-seal **300** from such a non-elastomeric material the amount of gas that seat seal **220** is exposed to can be eliminated or decreased. The u-seal **300** may be made from a non-permeable material and is located upstream from the seat seal **220**. When pressure is applied on the insides of the walls of the u-seal within the cup or cavity portion of the u-seal **300**, the walls of the u-seal **300** are pushed tight against the valve body **108** and the seat **208**. This results in a seal that is able to prevent or reduce gas and fluid flow between these components. In addition, since the u-seal **300** is formed from a non-permeable material, gas is unable to flow through the body of the u-seal **300** or permeate the u-seal **300**. By this configuration, the amount of gas, which may permeate through the seat seal **220** or that contacts the seat seal **220** is reduced or eliminated.

The u-seal backer **304** is an elastomeric or non-elastomeric component that is located adjacent to u-seal **300** and between the u-seal **300** and the valve body **108** toward the seat seal **220**. In some embodiments, the u-seal backer **304** provides additional structural support to prevent undesired deformation of the u-seal **300**. In some embodiments, the u-seal backer provides a further backup seal to prevent fluid and gas flow past the u-shaped backer **304**. In some embodiments, u-seal backer **304** is omitted.

There are many examples of the various embodiments described herein. A number of examples also follow.

Example 1. A dynamic slab gate valve comprising:

a valve body having a through bore through the length of the valve and formed with a gate cavity oriented transverse to and intersecting with the through bore;

a gate located within the gate cavity operable to move linearly in the direction of the gate cavity between an open position and a closed position;

a seat assembly disposed proximate the through bore, the seat assembly having a proximate side oriented toward the gate cavity so that the seat assembly contacts the gate when the gate is in the closed position and having a distal side on an opposite end from the proximate side;

the seat assembly comprising at least one seal, a seat, and a seat energizer, wherein the seat energizer is located on the distal side of the seat assembly and biases the seat assembly away toward the gate; and

wherein the seat assembly is sized and configured to translate in the direction of the through bore in response to pressure from the seat energizer.

Example 2. The dynamic slab gate valve of Example 1, further comprising at least one stem coupled to the gate wherein movement of the gate is actuated by movement of the at least one stem.

Example 3. The dynamic slab gate valve of Example 2, wherein the at least one stem comprises an upper stem and a lower stem wherein the upper stem is coupled to an upper side of the gate and the lower stem is coupled to a lower side of the gate.

Example 4. The dynamic slab gate valve of Example 3, wherein the gate can displace in the direction of the through bore relative to the upper stem and the lower stem in response to fluid pressures.

Example 5. The dynamic slab gate of Example 1, further comprising a lubricant pocket wherein a seal between the gate and the seat blocks fluid flow between the lubricant pocket and the through bore.

Example 6. The dynamic slab gate of Example 1, wherein the seat assembly further comprises at least one sediment blocker.

Example 7. The dynamic slab gate of Example 6, wherein the at least on sediment filter comprises a large sediment filter and a fine sediment filter.

Example 8. A dynamic slab gate valve, comprising:

a valve body having a through bore through the length of the valve and a gate cavity oriented transverse to and bisecting the through bore;

a gate located within the gate cavity operable to move in the direction of the gate cavity between an open position and a closed position;

a stem connected to a proximate end of the gate;

a lubricant pocket disposed within the valve body in fluid communication with at least a portion of the stem and a portion of the gate;

a skirt assembly at least partially disposed between the valve body proximate the gate cavity and the gate and at least partially disposed between the stem and the lubricant pocket;

wherein the skirt assembly comprises a skirt and a skirt biasing energizer;

wherein the skirt biasing energizer biases the skirt away from a portion of the valve body and toward the gate.

Example 9. The dynamic slab gate valve of Example 8, wherein movement of the gate is actuated by movement of the stem.

Example 10. The dynamic slab gate valve of Example 8, wherein the gate can displace in the direction of the through bore relative to the stem.

Example 11. The dynamic slab gate valve of Example 10, wherein a seal between the skirt plate and the gate prevents fluid communication between the through bore and the lubricant pocket.

Example 12. The dynamic slab gate valve of Example 11, wherein the skirt biasing energizer biases the skirt plate toward the gate to maintain the seal between the skirt plate and the gate.

Example 13. A dynamic seat assembly for a gate valve comprising:

at least one seal,

a seat, and

a biasing energizer,

wherein, when installed in a gate valve, the biasing energizer biases the seat assembly toward a gate of the gate valve.

Example 14. The dynamic seat assembly of Example 13, wherein contact between the gate and the seat assembly reduces fluid communication between a through bore of the gate valve and a lubrication cavity of the gate valve.

Example 15. The dynamic seat assembly of Example 13, further comprising at least one sediment filter.

Example 16. The dynamic seat assembly of Example 15, wherein the at least one sediment filter comprises a large sediment filter and a fine sediment filter.

Example 17. A skirt assembly for a gate valve, comprising:

a skirt plate, and

a skirt energizer coupled to one side of the skirt plate,

wherein, when installed in a gate valve, the skirt energizer biases the skirt plate toward a gate of a gate valve.

Example 18. The skirt assembly of Example 17, wherein the skirt assembly further comprises at least one lubricant guide coupled to the one side of the skirt plate.

Example 19. The skirt assembly of Example 17, wherein the skirt biasing energizer biases the skirt plate toward the gate when the gate translates in the direction of a through bore of the gate valve in response to fluid pressure.

Example 20. A dynamic slab gate valve comprising:

a valve body having a through bore through the length of the valve and a gate cavity oriented transverse to and intersecting with the through bore;

a gate located within the gate cavity operable to move in the direction of the gate cavity between an open position and a closed position;

a seat assembly disposed proximate the through bore, the seat assembly having an inward side oriented toward the gate cavity so that the seat assembly contacts the gate when the gate is in the closed position;

wherein the seat assembly comprises at least one seal, a seat, and a seat biasing energizer, and wherein the seat biasing energizer is located on an outboard side of the seat assembly and biases the seat assembly away toward the gate;

wherein the seat assembly is configured to translate in the direction of the through bore in response to pressure from the biasing energizer.

a stem connected to a proximate end of the gate;

a lubricant pocket disposed within the valve body in fluid communication with at least a portion of the stem and a portion of the gate;

a skirt assembly at least partially disposed between the valve body and the gate and at least partially disposed between the stem and the lubricant pocket;

wherein the skirt assembly comprises a skirt and a skirt biasing energizer;

wherein the skirt biasing energizer biases the skirt toward the gate.

Example 21. A dynamic slab gate valve comprising:

a valve body formed with a through bore and a gate cavity orthogonal to the through bore;

a gate disposed within the gave cavity, wherein the gate has a slab portion for halting flow in the through bore and an aperture portion for allowing flow through the gate valve and onward through the through bore; and

one or more dynamic features associated with the gate.

Example 22. The dynamic slab gate valve of Example 21, wherein the one or more dynamic features comprises a dynamic valve skirt.

Example 23. The dynamic slab gate valve of Example 21, wherein the one or more dynamic features comprises a dynamic seat assembly.

Example 23. The dynamic slab gate valve of Example 21, wherein the one or more dynamic features comprises a dynamic valve skirt and a dynamic seat assembly.

Example 24. The dynamic slab gate valve of Examples 1-7, 13-16, or 20-22, wherein the seat assembly further comprises a non-elastomeric u-seal, wherein the non-elastomeric u-seal is formed with a cross-sectional profile with a u or cup shape, and wherein the u-seal is disposed between the valve body and the seat and oriented so that fluid or pressure flow flows into the u or cup shape of the non-elastomeric u-seal.

Example 25. The dynamic slab gate valve of Example 23, wherein a wall portion of the non-elastomeric u-seal deforms to expand the u or cup shape of the non-elastomeric u-seal in response to fluid or gas pressure exerted within the u or cup shape of the non-elastomeric u-seal.

Example 26. The dynamic slab gate valve of Example 23, wherein the non-elastomeric u-seal is formed from PEET.

Example 27. The dynamic slab gate valve of Example 23, wherein the non-elastomeric u-seal is formed from PTFE.

Although the present invention and its advantages have been disclosed in the context of certain illustrative, non-limiting embodiments, it should be understood that various changes, substitutions, permutations, and alterations can be made without departing from the scope of the invention as defined by the claims. It will be appreciated that any feature that is described in a connection to any one embodiment may also be applicable to any other embodiment.

What is claimed:

1. A slab gate valve comprising:

a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction, which is orthogonal to the first direction;

a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side;

wherein the gate cavity is sized and configured to receive the slab gate;

wherein the first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position;

wherein the second portion of the slab gate valve is formed with a flow aperture for allowing flow there-through when the slab gate is in an open position;

an upper stem coupled to the slab gate for selectively moving the slab gate between the open position and the closed position;

a lower stem coupled to the slab gate and at least partially disposed within the gate cavity;

a downstroke lubricant cavity formed in a portion of the valve body proximate to the gate cavity for containing a lubricant for lubricating the lower stem in a down-stroke;

a first seat cavity formed in the valve body proximate a first intersection of the through-bore and the gate cavity on a first side of the first intersection;

a second seat cavity formed in the valve body proximate a second intersection of the through-bore and the gate cavity on a second side of the second intersection;

a first dynamic seat assembly comprising a first dynamic seat energizer disposed in the first seat cavity;

a second dynamic seat assembly comprising a second dynamic seat energizer disposed in the second seat cavity;

a first dynamic skirt, coupled to a first dynamic skirt energizer, disposed adjacent to the first side of the slab gate and disposed between the first seat cavity and the downstroke lubricant cavity, wherein the first dynamic skirt contacts the slab gate and the first dynamic skirt energizer contacts the valve body;

a second dynamic skirt, coupled to a second dynamic skirt energizer, disposed adjacent to the second side of the slab gate and disposed between the second seat cavity and the downstroke lubricant cavity, wherein the second dynamic skirt contacts the slab gate and the second dynamic skirt energizer contacts the valve body; and

wherein the first dynamic seat energizer, second dynamic seat energizer, first dynamic skirt energizer, and second dynamic skirt energizer are different energizers.

2. The slab gate valve of claim **1**, wherein the first dynamic seat assembly comprises:

a first cylindrical seat having a first side and a second side, wherein the second side is adjacent to the first side of the slab gate;

wherein a first debris ring slot is formed on an inner portion of the first cylindrical seat;

a debris ring proximate the first side of the first cylindrical seat and disposed at least partially within the first debris ring slot;

wherein the first cylindrical seat is formed with a seat energizer cavity proximate the first side of the first cylindrical seat; and

wherein the first dynamic seat energizer is disposed at least partially in the energizer cavity and operable to urge the first cylindrical seat toward the slab gate.

3. The slab gate valve of claim **2**, wherein the debris ring is perforated.

4. The slab gate valve of claim **2**, wherein the first dynamic seat assembly further comprises a first seat seal disposed within a seat seal cavity located on a perimeter of the first cylindrical seat, wherein the first seat seal provides a seal between the first dynamic seat assembly and the valve body.

5. The slab gate valve of claim **1**, wherein the first dynamic skirt energizer urges the first dynamic skirt toward the gate and wherein the second dynamic skirt energizer urges the second dynamic skirt toward the gate.

6. The slab gate valve of claim **5**, wherein the first dynamic skirt and the second dynamic skirt each further comprise a lubricant guide.

7. The slab gate valve of claim **5**,

wherein the first dynamic skirt has a first arcuate cutout that contacts the first dynamic seat assembly and wherein the first arcuate cutout conforms with a perimeter of the first dynamic seat assembly, and wherein the second dynamic skirt has a second arcuate cutout that contacts the second dynamic seat assembly and wherein the second arcuate cutout conforms with a perimeter of the second dynamic seat assembly.

8. The slab gate valve of claim **1**, wherein the upper stem coupled to the slab gate is coupled to the slab gate by a tee and t-slot connection.

9. The slab gate valve of claim **5**, wherein the first dynamic skirt energizer or the second dynamic skirt energizer has a longitudinal length that extends in a same direction as a longitudinal length of the slab gate.

10. The slab gate valve of claim **5**, wherein the first dynamic skirt energizer or the second dynamic skirt energizer has a longitudinal length that extends in a direction that is perpendicular to a longitudinal length of the slab gate.

11. The slab gate valve of claim **1**, where each of the first dynamic seat assembly and second dynamic seat assembly comprises a seat and a non-elastomeric u-seal, wherein the non-elastomeric u-seal is formed with a cross-sectional profile with a u or cup shape, and wherein the u-seal is disposed between the valve body and the seat and oriented so that fluid or pressure flow flows into the u or cup shape of the non-elastomeric u-seal.

12. The slab gate valve of claim **11**, wherein a wall portion of the non-elastomeric u-seal deforms to expand the u or cup shape of the non-elastomeric u-seal in response to fluid or gas pressure exerted within the u or cup shape of the non-elastomeric u-seal.

13. The slab gate valve of claim **11**, wherein the non-elastomeric u-seal is formed from PEET or PTFE.

14. A slab gate valve comprising:

a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction orthogonal to the first direction;

a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side;

wherein the gate cavity is sized and configured to receive the slab gate;

wherein the first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position;

wherein the second portion of the slab gate valve is formed with a flow aperture for allowing flow therethrough when the slab gate is in an open position;

an upper stem coupled to the slab gate for selectively moving the slab gate between the open position and the closed position;

a lower stem coupled to the slab gate and at least partially disposed within the gate cavity;

a downstroke lubricant cavity formed in a portion of the valve body proximate to the gate cavity for containing a lubricant for lubricating the lower stem in a downstroke;

at least one seat cavity formed in the valve body proximate an intersection of the through-bore and the gate cavity;

at least one dynamic seat assembly disposed in the at least one seat cavity;

wherein the at least one dynamic seat assembly comprises a dynamic seat energizer;

at least one dynamic skirt, coupled to a dynamic skirt energizer, disposed adjacent to the slab gate and disposed between the at least one seat cavity and the downstroke lubricant cavity, wherein the at least one dynamic skirt contacts the slab gate and the dynamic skirt energizer contacts the valve body; and

wherein the dynamic seat energizer is not the same energizer as the dynamic skirt energizer.

15. The slab gate valve of claim **14**, wherein the at least one dynamic seat assembly comprises:

a cylindrical seat having a first side and a second side, wherein the first side of the cylindrical seat is adjacent the slab gate;

a debris ring proximate the second side of the cylindrical seat;

wherein the cylindrical seat is formed with a seat energizer cavity proximate the second side of the cylindrical seat; and

the dynamic seat energizer is disposed in the seat energizer cavity and operable to urge the cylindrical seat toward the slab gate.

16. The slab gate of claim **14**,

wherein the at least one dynamic seat assembly comprises a first dynamic seat assembly and a second dynamic seat assembly;

wherein the at least one seat cavity comprises a first seat cavity and a second seat cavity, wherein the first seat cavity is adjacent to the first side of the slab gate and the second seat cavity is adjacent to the second side of the slab gate;

wherein the first dynamic seat assembly is disposed within the first seat cavity and the second dynamic seat assembly is disposed within the second seat cavity;

wherein the first dynamic seat assembly and the second dynamic seat assembly each comprise:

a cylindrical seat, having a first side and a second side, wherein the first side is adjacent the slab gate,

a debris ring proximate the second side, and

a seat energizer cavity proximate the second side;

wherein the dynamic seat energizer of the first dynamic seat assembly is disposed in the seat energizer cavity of the first dynamic seat assembly and is operable to urge the cylindrical seat of the first dynamic seat assembly toward the slab gate; and

wherein the dynamic seat energizer of the second dynamic seat assembly is disposed in the seat energizer cavity of the second dynamic seat assembly and is operable to urge the cylindrical seat of the second dynamic seat assembly toward the slab gate.

17. The slab gate valve of claim **16**, wherein the debris ring is perforated.

18. The slab gate valve of claim **15**, wherein the at least one dynamic seat assembly further comprises a seat seal disposed within a seat seal channel located proximate to a perimeter of the cylindrical seat, wherein the seat seal provides a seal between the cylindrical seat and the valve body.

19. A slab gate valve comprising:

a valve body formed with a through-bore in a first direction and formed with a gate cavity in a second direction orthogonal to the first direction;

a slab gate having a first portion and a second portion, wherein the slab gate has a first side and a second side;

wherein the gate cavity is sized and configured to receive the slab gate;

wherein the first portion of the slab gate is sized and configured to occlude the through-bore when the slab gate is in a closed position;

wherein the second portion of the slab gate is formed with a flow aperture for allowing flow therethrough when the slab gate is in an open position;

an upper stem coupled to the slab gate for selectively moving the slab gate between the open position and the closed position;

a lower stem coupled to the slab gate and at least partially disposed within the gate cavity;

a downstroke lubricant cavity formed in a portion of the valve body proximate to the gate cavity for containing a lubricant for lubricating the lower stem in a downstroke;

at least one dynamic skirt assembly disposed adjacent to the slab gate and between the slab gate and the downstroke lubricant cavity, wherein a first side of the at least one dynamic skirt assembly contacts the slab gate and a second side of the at least one dynamic skirt assembly contacts the valve body;

at least one dynamic seat assembly disposed within a seat cavity formed in the valve body proximate an intersection of the through-bore and the gate cavity;

wherein the at least one dynamic skirt assembly comprises a skirt plate proximate to the slab gate;

wherein the at least one dynamic seat assembly comprises a cylindrical seat proximate to the slab gate; and

wherein the skirt plate and the cylindrical seat are independently urged toward the gate by a plurality of energizers, wherein the plurality of energizers comprises a first energizer disposed between the skirt plate and the valve body and a second energizer disposed between the cylindrical seat and the valve body.

20. The slab gate valve of claim **19**, wherein the at least one dynamic skirt assembly at least partially prevents fluid flow between the downstroke lubricant cavity and the flow aperture of the second portion of the slab gate when the slab gate is in the closed position.

21. The slab gate valve of claim **19**, wherein the first energizer is coupled to the skirt plate.

22. The slab gate valve of claim **21**, wherein the at least one dynamic skirt assembly comprises a first dynamic skirt assembly and a second dynamic skirt assembly wherein the first dynamic skirt assembly is disposed adjacent to the first side of the slab gate and the second dynamic skirt assembly is disposed adjacent to the second side of the slab gate.

23. The slab gate valve of claim **19**, wherein the at least one dynamic skirt assembly further comprises a lubricant guide.

24. The slab gate valve of claim **19**, wherein the upper stem is coupled to the slab gate is coupled to the slab gate by a tee and t-slot connection.

25. The slab gate valve of claim **21**,

wherein the skirt plate has a longitudinal length and a lateral length;

wherein, when assembled within the slab gate valve, the longitudinal length of the skirt plate is oriented parallel to the second direction;

wherein the first energizer has a longitudinal length and a lateral length;

wherein, when assembled within the slab gate valve, the longitudinal length of the first energizer is oriented parallel to the second direction.

26. The slab gate valve of claim **21**,

wherein the skirt plate has a longitudinal length and a lateral length;

wherein, when assembled within the slab gate valve, the longitudinal length of the skirt plate is oriented parallel to the second direction;

wherein the first energizer has a longitudinal length and a lateral length;

wherein, when assembled within the slab gate valve, the longitudinal length of the first energizer is oriented perpendicular to the second direction.

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